CSS

Animation

```html
<!doctype html>
<html lang="en">
  <head>
    <meta charset="UTF-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1.0" />
    <title>Animation</title>
    <style>
      .box {
        border: 2px solid black;
        background-color: brown;
        color: white;
        width: fit-content;
        animation-name: slide;
        animation-duration: 5s;
        animation-timing-function: ease-in-out;
        /* animation-delay: 5s; */
        animation-iteration-count: infinite;
      }
      .box:hover {
        animation-play-state: paused;
      }

      @keyframes slide {
        0% {
          transform: translateX(0);
          background-color: blue;
        }
        25% {
          background-color: orangered;
        }
        50% {
          /* transform: scale(1.5); */
          background-color: greenyellow;
        }
        75% {
          background-color: aqua;
        }
        100% {
          transform: translateX(100px);
          background-color: yellow;
        }
      }

      .loading {
```

```
        width: 100px;
        height: 100px;
        /* background-color: bisque; */
        border: 10px solid gray;
        border-top-color: blue;
        border-radius: 50%;
        animation-name: rotate;
        animation-duration: 2s;
        animation-iteration-count: infinite;
        animation-direction: alternate-reverse;

      }

      .loading:hover{
        animation-play-state: paused;
      }

      @keyframes rotate {
        from{
            transform: rotate(0deg) translateY(0);
        }
        to{
            transform: rotate(360deg) translateY(20px);
        }
      }
    </style>
  </head>
  <body>
    <h1>Animation</h1>

    <div class="loading"></div>

    <div class="box">
      <h4>Slide animation</h4>
    </div>
  </body>
</html>
```

Project link -: https://solestyle-colorlib.pages.dev/

Theory Notes

1. CSS Animation Basics

- **Animations** allow elements to change styles over time.
- Key properties:
  - `animation-name` → links to `@keyframes`.
  - `animation-duration` → how long the animation runs.
  - `animation-timing-function` → speed curve (`ease`, `linear`, `ease-in-out`).
  - `animation-delay` → wait before starting.
  - `animation-iteration-count` → number of repeats (`infinite` for looping).
  - `animation-direction` → `normal`, `reverse`, `alternate`, `alternate-reverse`.
  - `animation-play-state` → `running` or `paused`.

2. @keyframes

- Defines the animation sequence.
- Syntax:

```
css
@keyframes slide {
  0% { transform: translateX(0); }
  100% { transform: translateX(100px); }
}
```

- Percentages (`0%`, `25%`, `50%`, etc.) represent stages of the animation.

3. Hover & Control

- `:hover` can pause or trigger animations.
- Example:

```
css
.box:hover {
  animation-play-state: paused;
}
```

4. Common Animation Effects

- **Translate** → move element (`translateX`, `translateY`).
- **Rotate** → spin element (`rotate(360deg)`).
- **Scale** → resize element (`scale(1.5)`).
- **Opacity** → fade in/out.
- **Color change** → background or text color transitions.

5. Differences: Transition vs Animation

- **Transition** → smooth change between two states (triggered by event like hover).
- **Animation** → continuous or multi-step changes defined in `@keyframes`.

Practice Tasks

Short Answer

1. What is the difference between `animation-duration` and `animation-delay`?
2. How does `animation-iteration-count: infinite` behave?
3. What does `animation-play-state: paused` do?
4. Why use `@keyframes` instead of `transition`?

Coding Exercises

1. Create a bouncing ball animation using `translateY`.
2. Animate a box that rotates 360° continuously.
3. Make text fade in and fade out using `opacity`.
4. Create a loading spinner with `border` and `rotate`.
5. Animate a button so it grows larger when hovered and shrinks back when not hovered.
6. Build a rainbow background animation that cycles through colors.
7. Add `alternate` direction so an element moves back and forth instead of restarting.